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Scan Paths

This article explains how to view Scan Paths based on respondent eye tracking data.

Scan paths allow visualization of the gaze path of individual respondents and their eye tracking data.

Scan paths can be overlaid onto static image or video replays.

Viewing Scan Paths in Replay Windows

- Scan paths can be found in any [Respondent Replay Window](#) for a study or analysis that contains eye tracking data.
- Scan Paths show the visual path for each respondent recording with Fixations (circles) and Saccades (lines).
 - Scan paths results from the selected [Fixation Classifier](#) - in other words, scan paths show fixation-classified gaze data

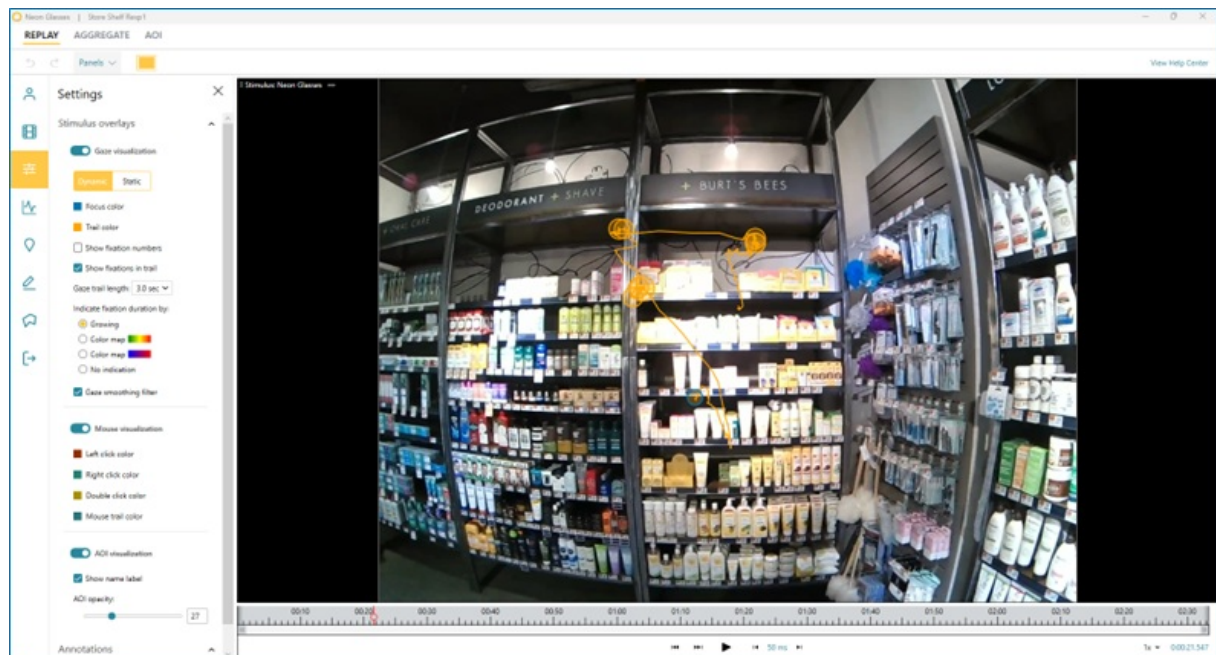
Visualization Settings Menu

Visualization **Settings menu** can be found in any Replay Window (along the left). These Settings control the Scan Path that appears in the stimulus panel.

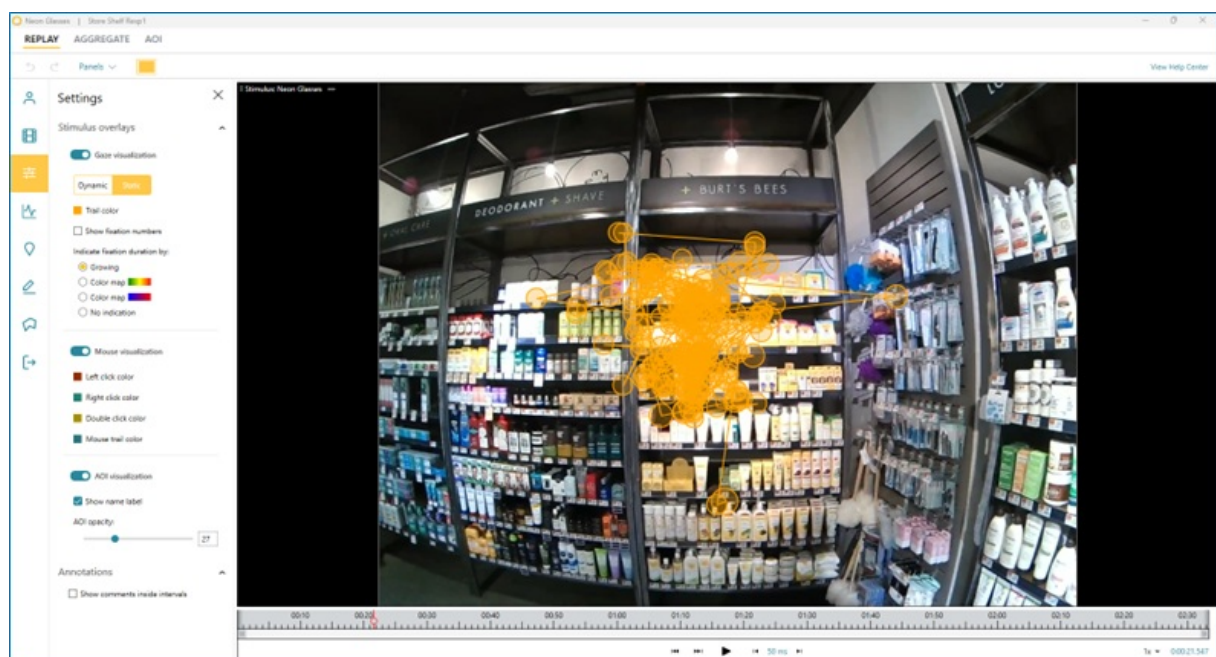
- **Static Scan Paths** overlays all eye tracking data for the stimulus, showing the visual path for the entire recording.
- **Dynamic Scan Path** overlays the eye tracking data according to the 'Gaze trail length' setting, showing the visual path for that window.

The **Export menu** can then be used to create export files:

- The Video and Image options create images (.png) or videos (.wmv, .mp4), as they appear in the Replay window.
 - Additional information about [Video and Image Exports from Replays](#)

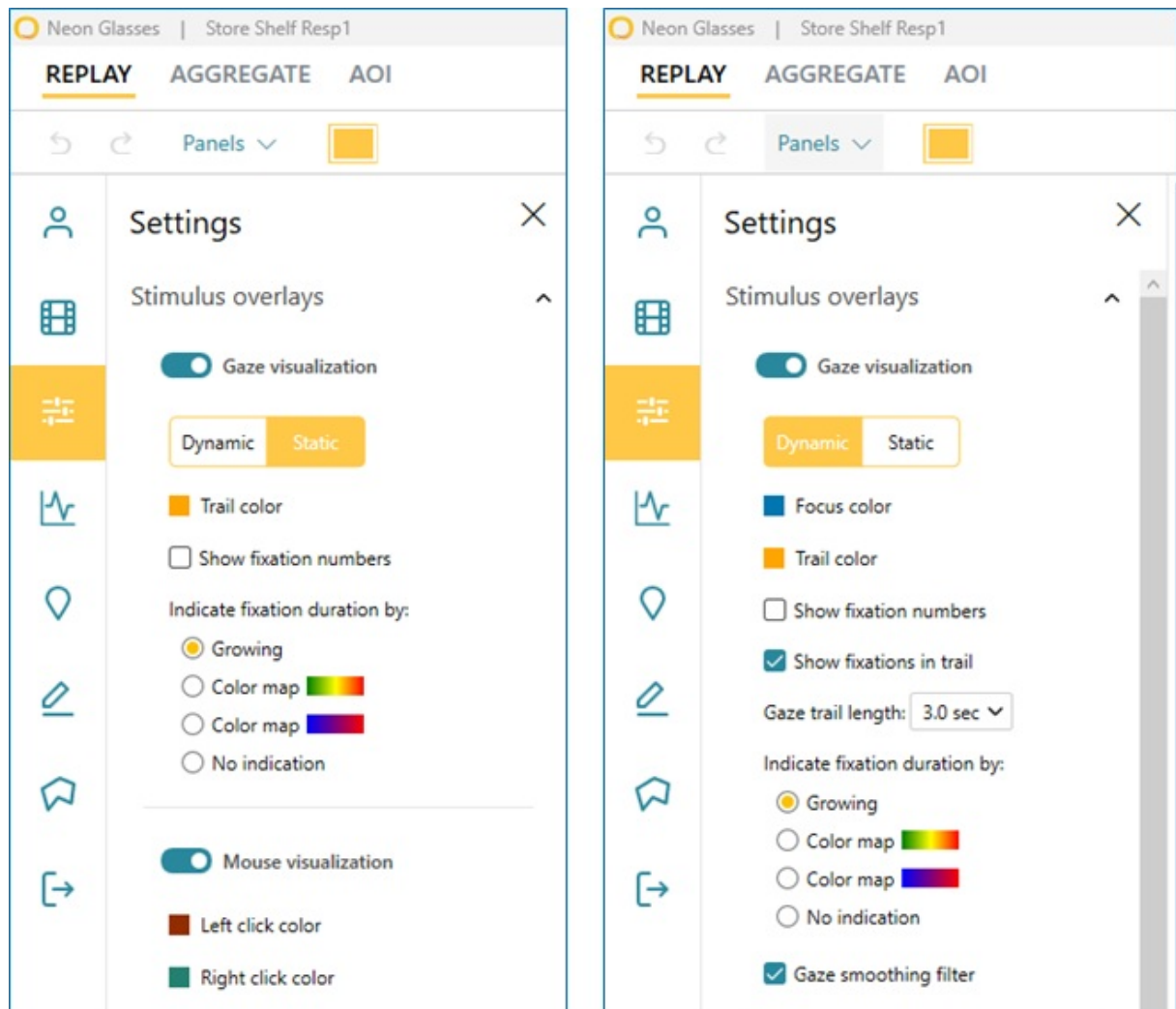


Example Dynamic Scan Path showing 3 sec Gaze trail length



Example Static Scan Path showing all eye tracking in the recording

Scan Path Visualization Options



Scan Path Settings Menus in Respondent Replay Window

Static Scan Paths Options

Trail color	Choose the color of the trail (saccades)
Show fixation numbers	Choose how to visualize the <i>order</i> of the fixations
Indicate fixation duration by	Choose how to visualize the <i>duration</i> of the fixations: by the size of the circles, or by color (red-green; blue-red)

Adjusting Static Scan Paths

- Static Scan Paths can be adjusted to specific intervals by creating [Scenes](#) that align with the desired interval/scan path.

- Alternatively, you can use a Dynamic Scan Path by navigating to the end of the interval and adjusting the 'Gaze trail length'. This approach is necessary for static stimulus types (e.g., images, survey slides, or 'gaze mapped images') can not be segmented into Scenes.

Dynamic Scan Paths Options

Focus color	Choose the color of the circles (fixations)
Trail color	Choose the color of the trail (saccades)
Show fixation numbers	Choose how to visualize the <i>order</i> of the fixations
Show fixations in trail	?
Gaze trail length	Choose how long to visualize the fixations (0 - 10 sec.)
Indicate fixation duration by	Choose how to visualize the <i>duration</i> of the fixations: by the size of the circles, or by color (red-green; blue-red)
Gaze smoothing filter	Choose to apply a gaze smoothing filter

Gaze smoothing filter

Gaze smoothing filter can be applied to mitigate the effects of signal noise in eye tracking data. This filter only applies to Scan Path visualization (and does not affect data or results).

Gaze Smoothing Filter Method

A Hamming window is applied to gaze data such that neighboring samples (gaze data closest in time) is averaged into the visualized/displayed gaze point to provide a smoother scan path. This smoothing filter is applied at time (t) along the gaze path:

1. The neighboring gaze points lying in a window of $[t - 80\text{ ms}, t + 80\text{ ms}]$ around the current points are selected. If missing gaze points are present in this interval, fewer points are considered.
2. Weighted averages are computed for the x and y coordinates of all points in the window, using a Hamming window of length 160 ms as weights.
3. The result of the weighted average gives the (x,y) coordinates of the smoothed point.

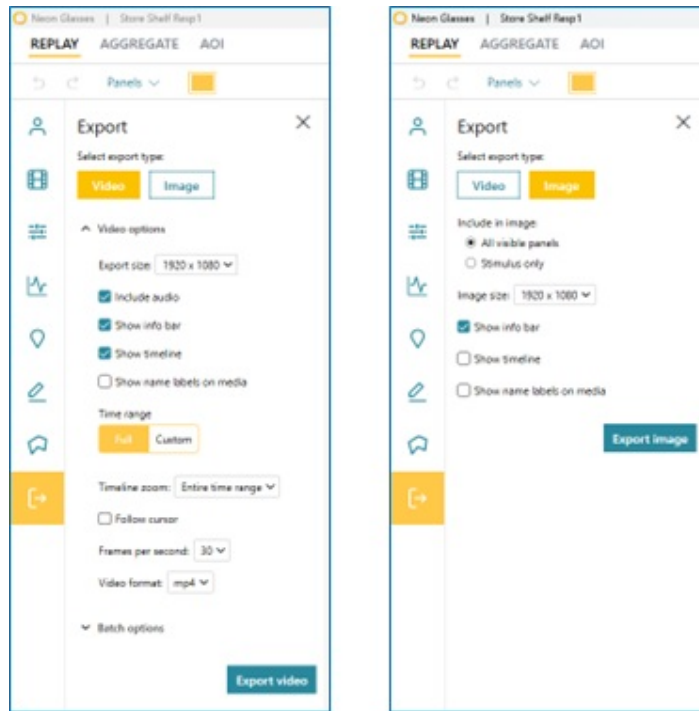


Scan Path with Gaze smoothing filter

Export Scan Paths

Scan Paths can be exported from any Replay window with the Export menu as video files (.mp4, .wmv) or image files (.png).

For information about these steps, refer to [Export: Images and Videos from Replays](#)



Export Menus